How to select a function randomly from a hash family?

→ UNIVERSAL HASH FAMILY

$h : U \to [m]$
hash function

U = universe of elements
m = given positive integer
$[m] = \{0, 1, 2, \ldots, m-1\}$

$\mathcal{H}$ is a universal hash family if

$\forall k_1, k_2 \in U, \quad k_1 \neq k_2 :$

$$\# h \in \mathcal{H} \text{ have } \underbrace{h(k_1) = h(k_2)}_{\text{collision}} \text{ is } \boxed{\frac{|\mathcal{H}|}{m}}$$

<u>Prop</u> $\Pr(\;) = \Pr_{h \in \mathcal{H}}(\;)$

$$\Pr(\text{collision}) = \frac{\# \text{BAD choices}}{\# \text{ choices}} = \frac{|\mathcal{H}|/m}{|\mathcal{H}|} = \boxed{\frac{1}{m}}$$

We actually build one example of $\mathcal{H}$:

$$h_{a,b}(x) = \big((a \cdot x + b) \,\%\, p\big) \,\%\, m$$

random choice (for a, b)

user specification (for p, m)

prime $p > m$

$$a \in \mathbb{Z}_p^{\times} = \{1, 2, \ldots, p-1\} \quad (\forall x \, \exists x^{-1} \text{ s.t. } x \cdot x^{-1} = 1)$$

$$b \in \mathbb{Z}_p = \{0, 1, \ldots, p-1\} = [p]$$

m, p : given

$$\mathcal{H} = \{ h_{a,b}(x) : a \in \mathbb{Z}_p^{\times},\ b \in \mathbb{Z}_p \}$$

example $m = 4$ $\quad p = 5$

$a \in \{1, 2, 3, 4\}$

$b = \{0, 1, 2, 3, 4\}$

$$\big((\underbrace{x+0}) \,\%\, 5\big) \,\%\, 4$$

$x+1, x+2, \ldots, 2x+0, 2x+1, \ldots, 2x+4$

$a=1 \quad b=0$

$$\big((3x+4) \,\%\, 5\big) \,\%\, 4$$

$a = 3 \quad b = 4$

<u>obs</u> $|\mathcal{H}| = \underbrace{(p-1)}_{\text{choices for } a}\ \underbrace{p}_{\text{choices for } b}$

<u>obs</u> choose $h \in \mathcal{H}$ uniformly at random

$\Updownarrow$

choose $a \in \mathbb{Z}_p^*$ and $b \in \mathbb{Z}_p$ uniformly at random

That is, choosing a random function here is equivalent to choosing a pair of random integers

Why this construction gives a <u>UNIVERSAL</u> HASH FAMILY?

① $|\mathcal{H}| = (p-1)p$

② $k_1, k_2 \in U$, $k_1 \neq k_2$

③ we should count for how many choices of a and b we have collision on k_1 and k_2

$$h_{ab}(k_1) = \underbrace{((ak_1+b)\%p)}_{r}\%m = r\%m$$

$$h_{ab}(k_2) = \underbrace{((ak_2+b)\%p)}_{s}\%m = s\%m$$

$$r \neq s$$

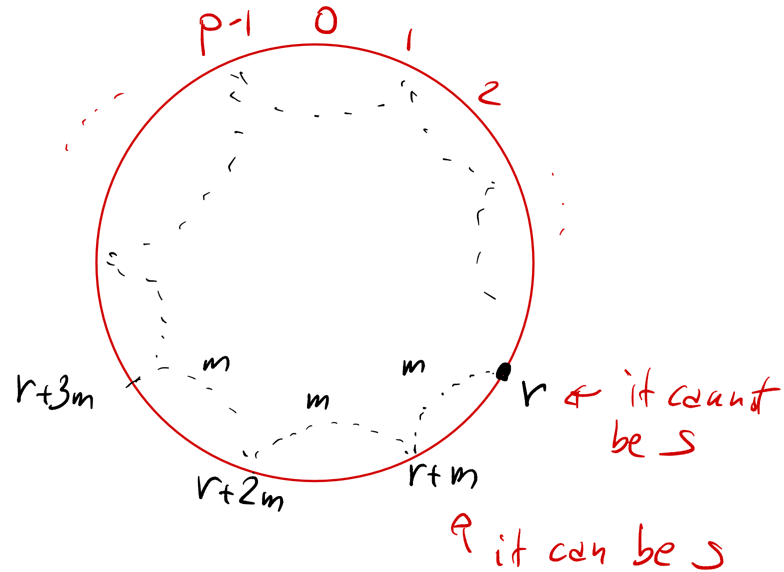
By h_p, there is a collision $\Leftrightarrow$ $r\%m = s\%m \Leftrightarrow |r-s| = 0 \mod m$

That is r and s differ by a multiple of m

3.a ✓ Fix r (we have p choices for r)

3.b - how many s can we have for r?

$$\underbrace{\left\lceil \frac{p}{m} \right\rceil}_{\substack{\text{how many}\\ \text{jumps of } m\\ \text{inside the circle}\\ \text{for choosing } s}} - \underbrace{1}_{\substack{\text{we exclude}\\ s=r}} \le \frac{p-1}{m}$$



Summing up:

#choices for a,b are #choices for r and s

$$\#D \le \underbrace{p}_{\substack{(3.a)\\ r}} \cdot \underbrace{\frac{p-1}{m}}_{\substack{s\\ (3.b)}} = \frac{p(p-1)}{m} = \frac{|H|}{m} \qquad \text{QED}$$

UNIVERSAL HASHING

$h(x) = (ax + b \% p) \% m$

p prime in $[m+1, 2m]$

$a, b \in [p]$, $a \neq 0$

PERFECT HASH

hash function $h : U \to [m]$ $|U| \gg m$

collisions $h(k_1) = h(k_2)$ with $k_1 \neq k_2$

What if you have knowledge of set $S \subseteq U$ to be stored? Positive answer when $m \geq |S|$

h is perfect for S if no two keys $k_1, k_2 \in S$ collide

G R S R R T

LAST FIRST

GROSSI ROBERTO

MARCO

MIRCO

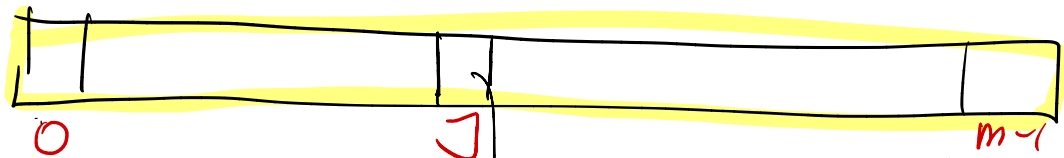
Sprugnoli

Perfect hash table(s)

$\mathcal{H}$ = universal family

1. Top level $\quad h \in \mathcal{H} \quad h : U \to [m]$

$m = 2n$

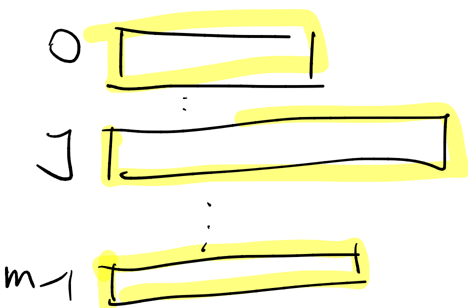


(here we accept collisions, but in a "controlled" way

n_j = # keys from S that are mapped to j

$= \big|\underbrace{\{k \in S : h(k) = j\}}_{S_j}\big|$

2. Bottom level



$h_j \in \mathcal{H}$

$m = n_j^2 = |S_j|^2$

[see Python code]

FIRST CLAIM (BOTTOM LEVEL)

$h_j : S_j \to [n_j^2]$, where $n_j = |S_j|$

$Pr(h_j \text{ is } \underline{\text{perfect}} \text{ for } S_j) \geq \frac{1}{2}$

[Before proving the claim ...
why $pr \geq \frac{1}{2}$ is good?]

random indicator variables to model collisions between any two keys k, ℓ

$$X_{k\ell} = \begin{cases} \boxed{1} & \text{if } k \neq \ell \text{ and } h_j(k) = h_j(\ell) \\ 0 & \text{otherwise} \end{cases}$$

$pr = \frac{1}{m}$, $m = n_j^2$

$$X = \sum_{k,l \in S_j} X_{kl} = \#\text{ collisions using } h_j \text{ on the keys in } S_j$$

$$X = 0 \iff \text{no collision}$$

Our claim $\iff$ $\Pr[X=0] = 1 - \underbrace{\Pr[X \geq 1]}_{< \frac{1}{2}} \geq \underbrace{\frac{1}{2}}_{\text{our goal}}$

$$\Pr[X \geq 1] \underset{a=1}{\leq} E[X] = \sum_{k,l} E[X_{kl}] = \sum_{k,l} \Pr[X_{kl}=1] = \underbrace{\sum_{k,l}}\frac{1}{n_j^2} = \binom{n_j}{2}\frac{1}{n_j^2}$$

$\boxed{\text{Markov's inequality}}$: $\Pr[X \geq a] \leq \frac{E[X]}{a}$ (proof in a future lecture)

$$\binom{n_j}{2}\frac{1}{n_j^2} = \frac{n_j(n_j-1)}{2n_j^2} < \frac{1}{2}$$

$\square$

SECOND CLAIM (SPACE)

$$\Pr_{h \in H}\Big(\underbrace{\sum_{j=0}^{m-1} n_j^2}_{X'} \le 4n\Big) \ge \frac{1}{2}$$

obs n_j depends on h

$$E[X'] = \sum_{j=0}^{m-1} E[n_j^2] = \sum_{j=0}^{m-1} E\left[n_j + 2\binom{n_j}{2}\right]$$

$$\left[\, a^2 = a + 2\binom{a}{2} \,\right] \quad a = n_j$$

$$= \underbrace{\sum_j E[n_j]}_{E[\sum_j n_j] = \underbrace{\,}_{n}} + 2\underbrace{\sum_j E\binom{n_j}{2}}_{\le \binom{n}{2}\frac{1}{m} = \frac{n-1}{2}\ \circledast} \le n + \cancel{2}\ \frac{(n-1)}{\cancel{2}}\ \frac{1}{2n} < 2n$$

$m = 2n$

$$\circledast \quad \sum_j E\left[\binom{n_j}{2}\right] = E\left[\sum_j \binom{n_j}{2}\right] = E[X] \quad \text{where } X = \sum_j \sum_{k,\ell \in S_j} X_{k\ell}$$

that is, all pairwise collision: x, y collide iff $\exists j$ s.t. $x, y \in S_j$

$$E[X] = \sum_j \sum_{k,\ell \in S_j} E[X_{k\ell}] = \sum_j \sum_{k,\ell \in S_j} \frac{1}{m} \leq \binom{n}{2} \frac{1}{m}$$

Summing up...

$$E[X'] < 2n$$

our claim $\Leftarrow$ $\Pr[X' > 4n] \leq \dfrac{E[X']}{4n} < \dfrac{2n}{4n} = \dfrac{1}{2}$ $\blacksquare$